
HARNESSING THE POWER OF CNNs FOR AUTOMATED SKIN CANCER CLASSIFICATION

Nathanael Wung
1009908325, nath.wung@mail.utoronto.ca
[Link to Google Colab](#)

Introduction

Skin cancer is among the most common malignancies worldwide, with over 80,000 cases reported annually in Canada [1]. Early detection significantly improves survival rates, reaching up to 99% when diagnosed at an early stage [2]. However, limited access to dermatologists—especially in rural areas—and the roughly 60% accuracy of visual inspection alone [3] highlight the need for automated, reliable diagnostic tools. This project applies deep learning, specifically convolutional neural networks (CNNs), to classify dermoscopic images from the HAM10000 dataset into seven skin lesion categories [4]. CNNs are well-suited for this task, as their hierarchical feature extraction captures color, texture, and border irregularities similar to those used by dermatologists [5]. This project represents an initial step toward building deployable tools that can be integrated into clinical workflows to enhance the efficiency of the diagnostic process.

Illustration

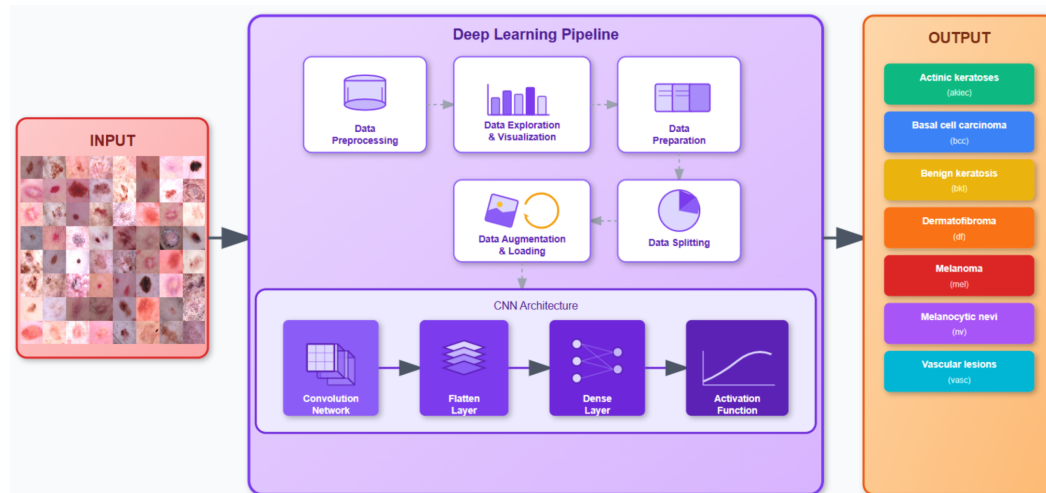


Figure 1: Pipeline for skin lesion classification.

Background & Related Work

Deep learning has become pivotal in skin lesion analysis, achieving dermatologist-level performance across multiple studies. The SLC-CNN combined preprocessing (augmentation, noise reduction, normalization) with SVM and XGBoost modules, attaining 99.6% classification and 95.3% segmentation accuracy on ISIC data [6]. A fusion of AlexNet, VGG16, and ResNet-18 features with an SVM classifier achieved AUCs up to 97.6% for seborrheic keratosis, highlighting the benefit of complementary CNN features [7]. The AIR-CNN integrated inception, residual, and attention blocks, reaching 91.6% accuracy on ISIC 2019 and reducing errors among visually similar lesions [8]. Fine-tuned ensembles of InceptionV3 and DenseNet-201 improved test accuracy from 79.9% to 88.5% [9], while the custom FCDS-CNN achieved 96% accuracy, outperforming larger pretrained models with greater efficiency [10].

Data Processing

The HAM10000 dataset consists of 10,015 dermoscopic images spanning seven skin lesion categories. The class distribution is shown in Figure 2. Data cleaning involved removing corrupted, duplicate, and incomplete samples to ensure high-quality inputs. All images were resized to 224×224 pixels and normalized to ensure consistent input for the model. The held-out 15% test set, containing unseen and unaugmented images, was reserved for unbiased performance evaluation. To further assess model generalization, a small set of external images not included in HAM10000 will be used for final testing. Since the nv class dominated the dataset, leading to class imbalance and reduced accuracy on rarer lesions, minority classes were oversampled using data augmentation, resulting in a balanced distribution as displayed in Figure 3. One key challenge during preprocessing was managing class imbalance without introducing overfitting, as excessive augmentation risked generating visually redundant samples.

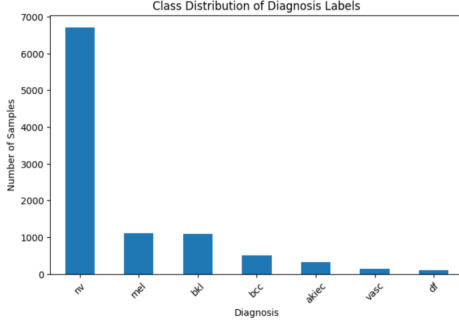


Figure 2: Distribution of skin lesion classes in the HAM10000 dataset.

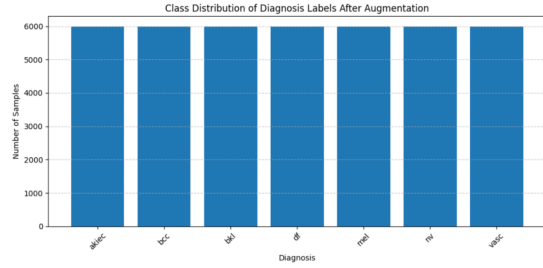


Figure 3: Distribution of skin lesion classes after augmentation.

Architecture

A pre-trained ResNet-18 CNN served as the final architecture. ResNet-18 uses skip connections between residual blocks to mitigate vanishing gradients and support stable training of deeper networks [11]. The encoder begins with a 7×7 convolution (stride 2), batch normalization, ReLU activation, and 3×3 max pooling, followed by four residual stages containing two 3×3 convolutional layers each. The number of filters increases from 64 to 512, enabling the extraction of progressively complex lesion features such as color variation, texture, and border irregularities. To adapt the model to the seven HAM10000 classes, the original fully connected layer was replaced with a new linear layer with seven output units. All layers up to layer3 were frozen, and only the final residual block (layer4) and the new classification head were fine-tuned. After the final convolutional stage, global average pooling feeds into the linear classifier. Training used CrossEntropyLoss with the Adam optimizer (learning rate = 1e-4, weight decay = 1e-4), along with a step learning-rate scheduler and early stopping to balance accuracy and generalization.

Baseline Model

A lightweight CNN was implemented as the baseline model to provide an initial benchmark for skin cancer classification. The architecture follows a simple encoder-decoder structure: the encoder uses three convolutional layers with channel sizes 32, 64, and 128, each followed by batch normalization, ReLU activation, and 2×2 max pooling to progressively extract low-level features. The decoder flattens the output and applies two fully connected layers to produce a seven-class probability distribution. This model contains relatively few parameters, requires minimal tuning, and can be reproduced easily, making it an appropriate baseline for comparison with deeper architectures such as ResNet-18. Training the baseline on the full HAM10000 dataset was computationally infeasible, so a subset of 1,200 training and 300 validation images was used. Even with limited data, the model learned meaningful patterns, achieving a best training accuracy of 44.2% and a best validation accuracy of 67.0%, providing a clear performance reference for evaluating improvements from transfer learning.

Quantitative Results

Building on the baseline results, the fine-tuned ResNet-18 model was evaluated to quantify the improvements gained from transfer learning. The model achieved best training and validation accuracies of 93.0% and 80.5%, respectively, as shown in Figure 4, demonstrating strong learning of discriminative features and reliable performance on the validation dataset. Class-level behaviour is summarized in Figure 5. The model performs particularly well on the dominant nv class, correctly identifying 617 samples, and also shows solid recognition for bcc and bkl. In contrast, classes with fewer examples—such as akiec and df—exhibit higher misclassification rates, reflecting the imbalance present in the validation set. Further detail is provided by Figure 6, which reports precision and recall for each lesion type. High recall for nv (0.91) and bcc (0.72) indicates strong sensitivity, while bkl achieves the highest precision (0.75), reflecting good specificity. Lower recall for akiec and mel highlights the difficulty of detecting underrepresented and visually subtle lesions. Overall, the model outperforms the baseline by leveraging transfer learning to learn robust and clinically meaningful features. Although performance on rare classes remains limited by data scarcity, the model demonstrates strong accuracy and reliable behaviour across the dominant lesion types.

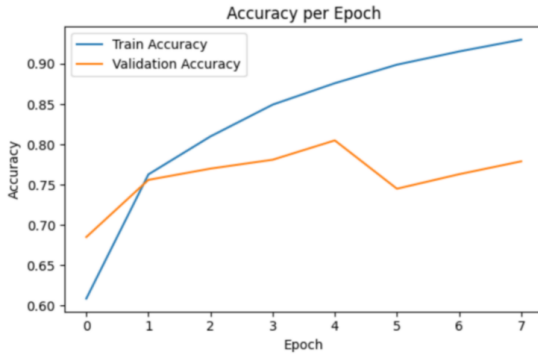


Figure 4: Accuracy curve of the fine-tuned ResNet-18 model.

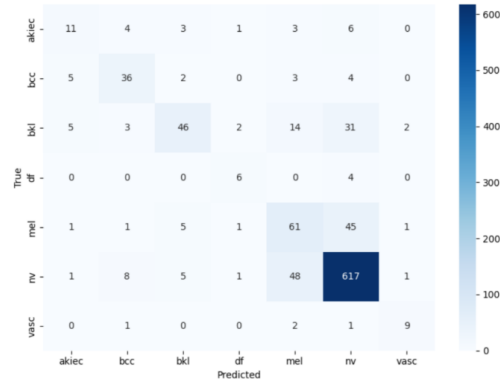


Figure 5: Confusion matrix of the fine-tuned ResNet-18 model.

	precision	recall
akiec	0.48	0.39
bcc	0.68	0.72
bkl	0.75	0.45
df	0.55	0.60
mel	0.47	0.53
nv	0.87	0.91
vasc	0.69	0.69

Figure 6: Classification report of the fine-tuned ResNet-18 model.

Qualitative Results

To complement the quantitative analysis, Figure 7 presents representative sample predictions that highlight the model's characteristic strengths and weaknesses. The model shows consistently correct predictions for nv, aligning with its strong quantitative performance, while errors involving classes such as bkl and mel reveal how subtle texture and color variations can lead to confusion among visually similar lesions. Model interpretability is illustrated through the Grad-CAM visualization in Figure 8. The heatmap shows that the model focuses primarily on the lesion region during prediction, indicating that it attends to clinically relevant structures rather than background artifacts. This provides reassurance that even when misclassifications occur, the model is leveraging meaningful visual cues. Together, these qualitative examples clarify why the model excels on common lesion types and struggles with rare or visually ambiguous cases, providing valuable context for the quantitative findings.

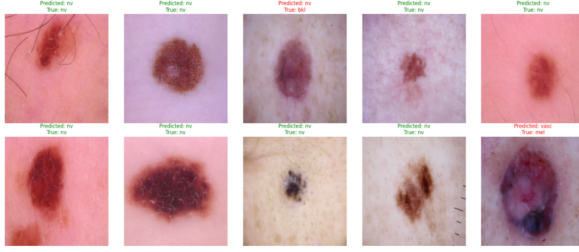


Figure 7: Predicted and true labels of the fine-tuned ResNet-18 model.

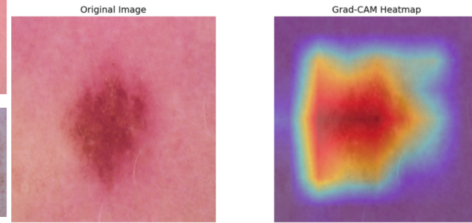


Figure 8: Grad-CAM visualization of the fine-tuned ResNet-18 model.

Evaluation on New Data

To assess the generalization ability of the model, two sources of unseen data were used: the held-out 15% test split from HAM10000 and a subset of images from the external BCN20000 dataset. Neither split was used during training, ensuring a fair generalization test. On the HAM10000 test set, the model achieved an accuracy of 77.2%, closely matching the validation accuracy of 80.5%, indicating strong generalization to in-distribution samples. To further evaluate robustness, the model was tested on BCN20000, a larger and more diverse dermoscopic dataset with different imaging conditions and lesion characteristics. The model achieved an accuracy of 73.6%, demonstrating that it transfers reasonably well to data outside the original dataset despite differences in acquisition settings and class distributions. These results show reliable performance on unseen data across differing datasets.

Discussion

The results show that the model performs well overall. Strong training and validation accuracies, along with consistent performance on both the held-out test set and the external BCN20000 dataset, indicate that the model learned generalizable features rather than overfitting to the training data. Grad-CAM further confirms that the model’s decisions rely on meaningful cues. A notable pattern is its strong performance on nv and difficulty with underrepresented categories such as vasc and df—an expected outcome of the skewed training distribution, though the model’s reliance on nv-like features was greater than anticipated. Qualitative examples show how subtle color and texture similarities lead to confusion among visually similar lesions, highlighting the limitations of heavily imbalanced datasets. Stable accuracy on BCN20000, despite differences in imaging devices and color profiles, suggests that domain shift remains a challenge but that transfer learning provides a solid foundation when applied to external data. This project also underscored that performance in dermoscopic classification is influenced not only by architecture but by dataset composition, annotation quality, and inherent visual ambiguity between lesion types. Overall, transfer learning strengthens classification, but achieving reliable performance on rare lesions requires mitigating imbalance and domain variability.

Ethical Considerations

Skin cancer classification raises significant ethical concerns, particularly regarding fairness and the consequences of misclassification. A key issue is that commonly used datasets like HAM10000 contain very few examples from patients with darker skin, reducing skin tone diversity and increasing the risk of poor accuracy for underrepresented groups [12]. Misclassification also carries serious impacts: false negatives may delay treatment, while false positives could lead to unnecessary procedures. These challenges underscore the need to assess model performance across diverse populations and to communicate limitations clearly so results are interpreted responsibly.

Project Difficulty & Quality

Skin cancer classification is inherently challenging due to high visual similarity between lesion types, severe class imbalance, and substantial variability in imaging conditions across datasets. Distinguishing subtle differences between benign and malignant lesions, particularly for underrepresented classes, makes this task far more complex than standard image classification problems. Despite these challenges, the model achieved strong and consistent performance across all evaluation settings: 80.5% on the validation dataset, 77.2% on the held-out test split, and 73.6% on the BCN20000 dataset. Maintaining this level of accuracy across datasets with different acquisition styles and class distributions demonstrates strong generalization and exceeds typical expectations for an introductory medical imaging model. The inclusion of external-dataset evaluation and interpretability analysis (Grad-CAM) further reflects the project’s depth, showing that the implemented approach effectively addressed the complexity of real-world dermoscopic image analysis.

References

- [1] Canadian Skin Cancer Foundation, “Skin Cancer,” *Canadian Skin Cancer Foundation*. [Online]. Available: <https://www.canadianskincancerfoundation.com/skin-cancer/>. [Accessed: Sep. 24, 2025].
- [2] The Skin Cancer Foundation, “Skin Cancer Facts & Statistics,” *The Skin Cancer Foundation*. [Online]. Available: <https://www.skincancer.org/skin-cancer-information/skin-cancer-facts/>. [Accessed: Sep. 24, 2025].
- [3] R. Marks, “An overview of skin cancers: incidence and causation,” *The Lancet Oncology*, vol. 3, no. 4, pp. 165–169, 2002. [Online]. Available: <https://www.thelancet.com/journals/lanonc/article/PIIS1470204502006794/abstract>. [Accessed: Sep. 24, 2025].
- [4] Harvard Dataverse, “HAM10000 dataset,” *Harvard Dataverse*. [Online]. Available: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T>. [Accessed: Sep. 24, 2025].
- [5] M. M. Musthafa, M. T. R., V. V. Kumar, and S. Guluwadi, “Enhanced skin cancer diagnosis using optimized CNN architecture and checkpoints for automated dermatological lesion classification,” *BMC Medical Imaging*, vol. 24, Art. no. 201, 2024. doi: 10.1186/s12880-024-01356-8. [Online]. Available: <https://bmcmmedimaging.biomedcentral.com/articles/10.1186/s12880-024-01356-8> (bmcmmedimaging.biomedcentral.com). [Accessed: Sep. 24, 2025].
- [6] U. Sara, “Automated Skin Cancer Classification and Detection Using SLCCNN with SVM and XG-Boost,” *Procedia Computer Science*, vol. 252, pp. 108–117, 2025. doi:10.1016/S1877050924034446. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1877050924034446>. [Accessed: Sep. 24, 2025].
- [7] A. Mahbod, G. Schaefer, C. Wang, R. Ecker, and I. Ellinger, “Skin Lesion Classification Using Hybrid Deep Neural Networks,” *arXiv preprint arXiv:1702.08434*, 2017. [Online]. Available: <https://arxiv.org/pdf/1702.08434>. [Accessed: Sep. 24, 2025].
- [8] S. Younas, A. B. Sargano, L. You, and Z. Habib, “Attention-Based Inception-Residual CNN: Skin Cancer Diagnosis with Attention-Based Inception-Residual CNN Model,” *Information*, vol. 16, no. 2, Art. 120, 2025. doi:10.3390/info16020120. [Online]. Available: <https://www.mdpi.com/2078-2489/16/2/120>. [Accessed: Sep. 24, 2025].
- [9] A. Mahbod, G. Schaefer, C. Wang, R. Ecker, and I. Ellinger, “Skin lesion classification with deep convolutional neural networks,” *JMIR Dermatology*, vol. 1, no. 1, e18438, 2020. [Online]. Available: <https://derma.jmir.org/2020/1/e18438/>. [Accessed: Sep. 24, 2025].
- [10] K. Nawaz, A. Zanib, I. Shabir, J. Li, Y. Wang, T. Mahmood, *et al.*, “Skin cancer detection using dermoscopic images with convolutional neural network,” *Scientific Reports*, vol. 15, Art. 7252, 2025. doi:10.1038/s41598-025-91446-6. [Online]. Available: <https://www.nature.com/articles/s41598-025-91446-6>. [Accessed: Sep. 24, 2025].
- [11] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” *arXiv preprint, arXiv:1512.03385*, 2015. [Online]. Available: <https://arXiv.org/pdf/1512.03385>. [Accessed: Sep. 24, 2025].
- [12] S. Morales-Forero, A. J. García-Arce, A. D. Quintero-Gil, and L. F. Durango, “An insight into racial bias in dermoscopy repositories: A HAM10000 data set analysis,” *JEADV Clinical Practice*, vol. 3, no. 3, pp. 836–843, 2024, doi:10.1002/jvc2.477. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/jvc2.477>. [Accessed: Sep. 24, 2025].